Rosh Ho

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Fullstack Software Engineer | Forward Deployed Engineer

Education

Columbia University **Sep 2021 - Aug 2025**

Computer Science *New York, New York*

- **Paper:** ReGlove arXiv:2512.11824 (IEEE BioRob 2026); Joint Inverse & Forward Dynamics arXiv:2503.12297
- **Awards:** Columbia DevFest Hackathon 2nd; Teacher College Mental Health x AI Hackathon 1st; UN Millennium Fellowship

Experience

Visentia AI **Jul 2025 - Current**

Forward Deployed Engineer, Fullstack Engineer *New York*

- Building an **agentic system** that explores and operates a legacy billing application (Medisoft), to automate claim lookup, status checks, and billing workflows without native API support with human-in-the-loop approval.
- Building an **agentic EMR platform**, building scheduling, documentation, and billing agents as independent, composable services behind a shared orchestration layer.
- Built a **two-stage Medicare claim denial prediction** pipeline (**XGBoost** + deterministic rules engine) over CMS claims data, reaching **74% PR-AUC** and cutting avoidable denials by **18%** across a 12-provider pilot.
- Engineered a **RAG appeal generation** system on **DeepSeek** over CMS LCD/NCD policies, with a **model-agnostic agent architecture** (swappable LLM backends) and prompt guardrails, cutting drafting time by **40%** across a 3-clinic pilot.
- Built role-based **React/Next.js** dashboards for physicians and RCM analysts with real-time denial risk visualizations (**Recharts**, **D3.js**).
- Architected an **end-to-end AWS pipeline**: **OCR ingestion**, relational schema design, **S3** archival, and **Lambda** orchestration, giving claims and appeals workflows an auditable data foundation.

LINE **May 2024 - Sep 2024**

Backend Software Engineering Intern *Taipei, Taiwan (ROC)*

- Contributed to a **high-performance Go backend** for dynamic ad placement on streaming platforms (AWS, MySQL) in a latency-sensitive production environment. • Engineered **high-performance backend** for dynamic ad placement on streaming platforms using **Go, AWS,** and **MySQL, reducing request latency by 13%** in a **latency-sensitive production environment**.
- Designed and implemented **automated unit-testing pipeline** in **Go**, improving **API reliability** and **scalability** for a system serving **millions of daily requests**.

AgGen Inc. **Jan 2022 - Aug 2024**

Fullstack Engineer (part-time) *Davis, California*

- Co-built an insect farming system converting organic waste into fertilizer and protein; **raised $65,000+ pre-seed funding**, won a **NYSERDA sustainability grant**, and partnered with Western Placer Waste Management Authority on pilot food-waste processing.
- Designed and built an **IoT monitoring and control system** for insect rearing units, using **microcontroller-based sensor nodes** (temperature, humidity, CO_2) streaming telemetry over **MQTT** to a **Python** backend and **React** dashboard, automating climate control and improving yield consistency by 23%.

Blackstone **May 2023 - Aug 2023**

Software Engineering Intern *New York, New York*

- Built a **Python** monitoring pipeline applying **anomaly detection** to data center sensor data, with **Grafana/Prometheus** dashboards and **Airflow**-automated reporting.

Singapore Armed Forces **Sep 2018 - Sep 2020**

Automation Engineer (Corporal) *Singapore, Singapore*

- Developed Python and Bash scripts to automate ETL pipelines over **SAP** data exports, replacing manual workflows across **8,000+** medical records with validated loads into a **MySQL** database.
- Built **UiPath RPA bots** with custom error-handling, retry logic, and **SQL stored procedures** to standardize reporting and reduce manual analysis time by **60%**.

Skills

Languages	Python, TypeScript, Go, C/C++, SQL
Technologies	React, Next.js, AWS (Lambda, S3, Bedrock), PyTorch, XGBoost, LangChain, MySQL, Linux, Git
Specialties	Agentic systems, RAG, legacy system automation, low-latency backends